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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB MANUAL

OOP THROUGH JAVA LAB

(R25)

B.TECH II YEAR – II SEM

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1. Vision, Mission & PEOs

Vision

To be a centre of excellence in Computer Science and Engineering education and research, producing globally competent professionals with strong ethical values, innovative mind-set, and commitment to societal development.

Mission

- To provide high-quality education in computer science and engineering, fostering analytical thinking, technical expertise, and innovation.
- To promote research, industry interaction, and hands-on learning to address real-world challenges and emerging technologies.
- To nurture ethically responsible, socially conscious, and globally competent professionals committed to lifelong learning and societal development.

PEO's

Program Educational Objectives(PEOs):	
PEO1	Build successful careers, pursue higher education or embark on entrepreneurial ventures by applying strong foundations in computer science, engineering principles, and modern technologies.
PEO2	Design and develop effective computing solutions to real-world problems through critical thinking, research aptitude, and continuous learning.
PEO3	Demonstrate ethical conduct, leadership, teamwork, and a commitment to sustainable development and societal well-being.

2. POs & PSOs

Program Outcomes of CSE Department (POs):	
PO1	**Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	**Design / Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal and environmentalconsiderations.
PO4	**Conduct investigations of complex problems**: using research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
PO5	**Modern Tool Usage**: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The Engineer and Society**: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	**Environment and Sustainability**: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional e t h i c s and responsibilities and norms of engineering practice.
PO9	**Individual and Team Work**: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large,

	such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
PO11	**Project Management and Finance**: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long Learning:** Recognize the need for and have the preparation and ability to engage in independent and life- long learning in the broadest context of technological change. Any signatory needs toprovide an overview of its learning outcomes and confirm that compliance of programs.

Program Specific Outcomes(PSOs):	
PSO1	Apply fundamental concepts of computer science to design, develop, and deploy reliable and efficient software systems using modern programming tools and methodologies.
PSO2	Analyze, model, and solve real-world problems using techniques in data science, artificial intelligence, and machine learning.
PSO3	Design and integrate computing systems with a focus on networking, cloud computing, Cyber Security, Blockchain, and IoT for secure and scalable applications.

3. COURSE OBJECTIVES & OUTCOMES

S.No	COURSE OBJECTIVES- The student will:
1	Learn the concepts of operators, control statements, type conversion, constructors and string handling
2	Understand the implementation of inheritance and polymorphism
3	Gain the knowledge of creation of user defined packages and interfaces
4	Familiar with exception handling, multithreading and event handling
5	Understand the concepts of applets, Swings in Java.

S.No	COURSE OUTCOMES- The student will be able to:
1	Implement arithmetic operators, control statements, type conversion, constructors and string handling.
2	Apply the OOP concepts using java.
3	Create user defined Packages and Interfaces
4	Implement Exception handling and Multithreading
5	Design GUI using Applets and AWT

4. LIST OF EXPERIMENTS

Experiment 1:

Write java programs that implement the following

a) Constructor b) Parameterized constructor

c) Method overloading d) Constructor overloading.

Experiment 2:

a) Write a Java program that checks whether a given string is a palindrome or not. Ex: MADAM is a palindrome.

b) Write a Java program for sorting a given list of names in ascending order.

c) Write a Java Program that reads a line of integers, and then displays each integer and the sum of all the integers (Use String Tokenizer class of java.util

Experiment 3:

Write java programs that uses the following keywords

a) this b) super

c) static d) final

Experiment 4:

a) Write a java program to implement method overriding

b) Write a java program to implement dynamic method dispatch.

c) Write a Java program to implement multiple inheritance.

d) Write a java program that uses access specifiers

Experiment 5:

Write a Java program that reads a file name from the user, then displays information about

whether the file exists, whether the file is readable, whether the file is writable, the type of fileand the length of the file in bytes.Write a Java program that reads a file and displays the file on the screen, with a line numberbefore each line.Write a Java program that displays the number of characters, lines and words in a text file.

Experiment 6:

a) Write a Java program for handling Checked Exceptions.

b) Write a Java program for handling Unchecked Exceptions

Experiment 7:

a) Write a Java program that creates three threads. First thread displays "Good Morning" every one second, the second thread displays "Hello" every two seconds and the third thread displays "Welcome" every three seconds.

b) Write a Java program that correctly implements producer consumer problem using the concept of inter thread communication

Experiment 8:

Write a Java program that works as asimple calculator. Use a grid layout to arrange button for the digits and for the +, -,*, % operations. Add a text field to display the result

Experiment 9:

a) Write a Java program for handling mouse events.

b) Write a Java program for handling key events.

Experiment 10:

Write a program that creates a user interface to perform integer divisions. The user nters two numbers in the text fields Num1 and Num 2.

The division of Num1 and Num2 is displayed in the Result field when the Divide button is clicked. If Num1or Num2 were not an integer, the program would throw Number Format Exception. If Num2 were Zero, the program would throw an Arithmetic Exception and display the exception in a message dialog box.

Experiment 11:

a) Write a java program that simulates traffic light. The program lets the user select one of three lights: red, yellow or green. When a radio button is selected, the light is turned on, and only one light can be on at a time No l Light is on when the program starts.

b) Write a Java program that allows the user to draw lines, rectangles and ovals

Experiment 12:

Create a table in Table.txt file such that the first line in the file is the header, and the remaining lines correspond to rows in the table. The elements are separated by commas. Write a java program to display the table using JTable component

PROGRAMS

EXPERIMENT-1

1.Writeajava program to implement the following:

a)Constructor b)parameterized constructor
c) Method overloading d)Constructor overloading

1a)Constructor

```
public class MyDefaultConstructor
{

public MyDefaultConstructor(){
System.out.println("Iam inside default constructor...");
}

public static void main(Stringa[])
{
MyDefaultConstructor mdc=new MyDefaultConstructor();
}
}
```

Output
Iam inside default constructor...

1b)Parameterized constructor

```
Public class MyParameterizedConstructor
{
Private Stringname;
public MyParameterizedConstructor(Stringstr)
{
this.name=str;
System.out.println("I am inside parameterized
constructor.");
System.out.println("The parametevalu eis:"+str);
}
public static void main(Stringa[])
{
MyParameterizedConstructor
mpc= new
MyParameterizedConstructor
("abcde");
}
}
```

Output:
I am inside parameterized constructor.The parameter value is: abcd

1c) Method overloading

```
class Calculator {
intadd(inta, intb) {
return a + b;
}
intadd(inta, intb, intc) {
return a + b + c;
}
double add(double a, double b) {
return a + b;
}
}
public class  Methodol{
public static void main(String[] args) {
Calculator c = new Calculator();
System.out.println(c.add(10, 20));
System.out.println(c.add(10, 20, 30));
System.out.println(c.add(2.5, 3.5));
}
}
```

Output:

30

60

6.0

1 d) Constructor overloading

```
class ConstructorOverload {
    int a, b;
    // Default constructor
    ConstructorOverload() {
        a = 0;
        b = 0;
    }
    // Constructor with one parameter
    ConstructorOverload(int x) {
        a = x;
        b = 0;
    }
    // Constructor with two parameters
    ConstructorOverload(int x, int y) {
        a = x;
        b = y;
    }

    void display() {
        System.out.println("a = " + a + ", b = " + b);
    }

    public static void main(String[] args) {
        ConstructorOverload obj1 = new ConstructorOverload();
        ConstructorOverload obj2 = new ConstructorOverload(10);
        ConstructorOverload obj3 = new ConstructorOverload(10, 20);
        obj1.display();
        obj2.display();
        obj3.display();
    }
}
```

Output

a = 0, b = 0
a = 10, b = 0
a = 10, b = 2

EXPERIMENT-2

2a) write a java program that checks whether a given string is a palindrome or not

Ex: MADAMisapalindrome

```
impor tjava.util.*;
class Palindrome
{
public static void main(Stringargs[])
{
String original,reverse= "";
Scanner in=new Scanner(System.in);
System.out.println("Enter a string to check if it is a palindrome");
original=in.nextLine();
intlength=original.length();
for ( inti = length - 1; i>= 0; i-- )
reverse = reverse + original.charAt(i);
if (original.equals(reverse))
System.out.println("Entered string is a palindrome.");
else
System.out.println("Entered string is not a palindrome.");
}
}
```

OUTPUT:

Enter a string to check if it is a palindrome
MADAM
Entered string isa palindrome

2b) Write a Java program for sorting a given list of names in ascending order.

```
import java.util.*;
class SortNames {
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    // Read number of names
    System.out.print("Enter number of names: ");
    int n = sc.nextInt();
    sc.nextLine(); // consume newline
    String[] names = new String[n];
    // Input names
    System.out.println("Enter the names:");
    for (int i = 0; i < n; i++) {
        names[i] = sc.nextLine();
    }
    // Sort names in ascending order
    Arrays.sort(names);
    // Display sorted names
    System.out.println("\nNames in ascending order:");
    for (String name : names) {
        System.out.println(name);
    }

    sc.close();
}
}
```

Output:

```
Enter number of names: 4
Enter the names:
Ravi
Anil
Sita
Balu

Names in ascending order:
Anil
Balu
Ravi
Sita
```

2.c)write a java program that reads a line of integers and then displays Each integer and the sum of all the integers use String Tokenizer class of java.util).

```
import java.util.StringTokenizer;
import java.util.Scanner;
class tokens
{
public static void main(String[]args)
{
Scanner input=new Scanner(System.in);
String sentence=input. nextLine();
String temp;
intk, total=0;
StringTokenizers1=new StringTokenizer(sentence);
System.out.println("Total Number of tokens:"+s1.countTokens());
while(s1.hasMoreTokens())
{
temp=s1.nextToken();
k=Integer.parseInt(temp);
total+=k;System.out.pr int(k+"\t");
}
System.out.println("Sum of tokens:"+total);
}
}
```

Output:

Integers are:
10
20
30
40

Sum of integers = 100

EXPERIMENT-3

3.Write java programs that uses the following keyword

a)this b)super

c)static d)final

3a) this

```
Student(int id, String name) {
this.id = id; // refers to current class instance variable
this.name = name;
}
void display() {
System.out.println("ID: " + id + ", Name: " + name);
}
}
public class ThisKeywordDemo {
public static void main(String[] args) {
Student s1 = new Student(101, "Ravi");
Student s2 = new Student(102, "Anil");
s1.display();
s2.display();
}}
```

OUTPUT

ID: 101, Name: Ravi

ID: 102, Name: Anil

3b)SUPER

```
class Person
{
   Person(int Age)
   {
     System.out.println("Person class Constructor");
    System.out.println("Person Age is“+ Age);
   }}
class Student extends Person
{
   Student()
   {
      super(25);
    System.out.println("Student class constructor");
   }}
class Test
{
   public static void main(String[] args)
   {
      Student s = new Student();
   }}
```

Output:

Person class Constructor

Person Age is 25

Student class constructor

3c)

STATIC

```
class StaticExample
{
  static double r=10;
  static double l=2.5,b=5.75,ar;
 void CircleArea()
{
   double ac;
   ac=3.14*r*r;
   System.out.println("Area of Circle:" +ac);
}
 static void RectArea()
{
   double ar;
   ar=l*b;
   System.out.println("Area of Rectangle:" +ar);
}
static void CirclePerimeter()
{
   double pc;
   pc=2*3.14*r;
   System.out.println("Perimeter of Circle:" +pc);
}
}

class Test1
{
public static void main(String args[])
{
    StaticExample obj=new StaticExample();
    obj.CircleArea();
    StaticExample.RectArea();
    StaticExample.CirclePerimeter();
}
}
```

Output:

Area of Circle: 314.0
Area of Rectangle: 14.375
Perimeter of Circle: 62.800000000000004

4 d)FINAL

```
    class FinalDemo {
    final int x = 10; // final variable
    void display() {
        // x = 20; ✕ Error: cannot change final variable
        System.out.println("Value of x: " + x);
    }
}

class TestFinal {
    public static void main(String[] args) {
        FinalDemo obj = new FinalDemo();
        obj.display();
    }
}
```

Output:

Value of x: 10

EXPERIMENT-4

4a) write a java program to implement method overriding

```
class A {
  int i, j;
  A(int a, int b)
{
    i = a;
    j = b;
  }
void show() {
    System.out.println("i and j: " + i + " " + j);
  }}
class B extends A {
  int k;
  B(int a, int b, int c) {
    super(a, b);
    k = c;
  }
void show()
{
    System.out.println("k: " + k);
  }
}
class Override {
  public static void main(String args[])
{
    B subOb = new B(1, 2, 3);
   subOb.show(); // this calls show() in B
  }
}
```

Output: k: 3

---

b) write a java program to implement dynamic method dispatch

```
class A {
  void callme() {
    System.out.println("Inside A's callme method");
  }
}
class B extends A {
  void callme() {
    System.out.println("Inside B's callme method");
  }
}
```

```
class C extends A {
  void callme() {
    System.out.println("Inside C's callme method");
  }
}
class Dispatch {
  public static void main(String args[]) {
    A a = new A(); // object of type A
    B b = new B(); // object of type B
    C c = new C(); // object of type C
     A r; // obtain a reference of type A
    r = a; // r refers to an A object
    r.callme(); // calls A's version of callme
    r = b; // r refers to a B object
    r.callme(); // calls B's version of callme
    r = c; // r refers to a C object
    r.callme(); // calls C's version of callme
  }
}
```

Output:

```
Inside A's callme method
Inside B's callme method
Inside C's callme method
```

c) write a java program to implement multiple inheritance

```
interface Sports
{
    void play();
}
interface Cultural {
    void dance();
}
interface Student extends Sports, Cultural
{
    void study();
}
// Class implementing child interface
 class Person implements Student {
    public void play() {
        System.out.println("Playing sports");
    }
public void dance() {
 System.out.println("Performing cultural dance");
   }
```

```
    public void study() {
        System.out.println("Studying subjects");
    }
}
// Main class
public class Mainclass {
    public static void main(String[] args) {
        Person obj = new Person();
        obj.play();
        obj.dance();
        obj.study();
    }
}
```

Output:

Playing sports
Performing cultural dance
Studying subjects

d) write a java program that uses access specifiers

```
class AccessDemo {
    private int a = 10;        // Private
    int b = 20;                // Default
    protected int c = 30;      // Protected
    public int d = 40;         // Public
void display() {
    System.out.println("Inside class:");
    System.out.println("Private a = " + a);
    System.out.println("Default b = " + b);
    System.out.println("Protected c = " + c);
    System.out.println("Public d = " + d);
  }
   }
   public class TestAccess {
      public static void main(String[] args) {
         AccessDemo obj = new AccessDemo();
         obj.display();
         System.out.println("\nAccessing from another class:");
         // System.out.println(obj.a); // ✕ Not accessible (private)
         System.out.println("Default b = " + obj.b);
         System.out.println("Protected c = " + obj.c);
         System.out.println("Public d = " + obj.d);
      }
```

Output:

```
Inside class:
Private a = 10
Default b = 20
Protected c = 30
Public d = 40

Accessing from another class:
Default b = 20
Protected c = 30
Public d = 40
```

EXPERIMENT-5

a .write a java program that reads a file name from the user ,file name from

The user,then displays information about whether the file exists,whether the file is readable,whether the file is writable,the type of file and the length of the file in bytes

```
import
java.util.Scanner; import
java.io.File;
class FileDemo
        {
        Public static void main(String[]args)
        {
        Scanner input=new Scanner(System.in);
        Strings=input.nextLine();
        File f1=new File(s);
        System.out.println("File Name:"+f1.getName());
        System.out.println("Path:"+f1.getPath());
        System.out.println("Abs Path:"+f1.getAbsolutePath());
        System.out.println("Parent:"+f1.getParent());
        System.out.println("This file is:"+(f1.exists()?"Exists":"Does not exists"));
        System.out.println("Isfile:"+f1.isFile());
        System.out.println("IsDirectory:"+f1.isDirectory());
        System.out.println("Is Readable:"+f1.canRead());
        System.out.println("IS Writable:"+f1.canWrite());
        System.out.println("IsAbsolute:"+f1.isAbsolute());
        System.out.println("File Last Modified:"+f1.lastModified());
        System.out.println("File Size:"+f1.length()+"bytes");
        System.out.println("IsHidden:"+f1.isHidden());
        }
        }
```

Output:

```
Fibonacci.java File
Name:Fibonacci.javaPath:
Fibonacci.java
Abs Path: c:\sameer\Fibonacci.java
Parent:Null
This file is:ExistsIsfile:true
IsDirectory:false
IsReadable:true
IsWritable:true
IsAbsolute:false
File Last Modified:1206324301937
FileSize: 406 bytes
IsHidden:false
```

b) write a java program that reads a file and displays the file on the screen with a linenumber before each line

```
import
java.util.*;
import
java.io.*;

public classTest{
public static void main(String []args)t hrows IOException
{
Scanner keyboard = new Scanner(System.in);
System.out.print("Enter a filename:");
String filename =keyboard.nextLine();

File file=new File(filename);
Scanner inputFile = new Scanner(file);
Stringline=inputFile.nextLine();
int lineNumber=0;
while(inputFile.hasNext())
{
String name = inputFile.nextLine();
System.out.println(lineNumber+ ":"+name);`
linenumber++;
System.out.println(name);
}
inputFile.close();

}

}
```

OUTPU
T:

```
Enter a file
name:a.txt 0 :abc
1:abcdefghij
2: 122345667899900
```

c) write a java program that displays the number of characters ,lines and words in a text file.

```
importjava.io.*;
class
wordcount
{
public static int
words=0; public static
int lines=0; public
static int chars=0;
public static void wc(InputStreamReader isr) throws IO Exception
{
int c=0;
Boolean
lastwhite=true;
while((c=isr.read())!=-1)
{
chars++;if(c=='\n')lines
++;
if(c=='\t'||c=='||c=='\n')
++words;if(chars!=0)
++chars;
} }
public static void main(String [] args)
{
FileReader fr;
try
{
if(args.length==0)
{
wc(new InputStreamReader(System.in));
}
else
{
for(int i=0;i<args.length;i++)
{
fr=new FileReader(args[i]);
wc(fr);
}}
}
catch(IOExceptionie)
{
return;
}
System.out.println(lines+""+words+""+chars);
}}
```

Output:

My name is Renuka
1 4 16

EXPERIMENT-6

a) writea java program for handling checked exceptions

```
 import java.io.*;
class CheckedExample {
  public static void main(String args[]) {
     try {
        FileReader fr = new FileReader("data.txt");
        BufferedReader br = new BufferedReader(fr);
        System.out.println(br.readLine());
     } catch (IOException e) {
        System.out.println("File error occurred");
     }
  }
}
```

Output:

If file exists (data.txt)
Hello World (or first line of the file)
If file does NOT exist
File error occurred

b) write a java program for handling unchecked exceptions

```
class UncheckedExample {
   public static void main(String args[]) {
      int a = 10;
      int b = 0;
      int c = a / b;  // ArithmeticException
      System.out.println(c);
   }
}
```

OUTPUT:
DividebyZero Exception:ArithmeticException

Note:
If you compile this code, it would compile successfully however when you will run it,it would throw ArithmeticException. That clearly shows that unchecked exceptions are not checked at compile-time, they are being checked at runtime.

EXPERIMENT-7

a) **write a java program that creates three threads. first thread displays "goodmorning" every one second,the second thread displays "hello" every two secondsandthe thread displays"welcome"every three seconds.**

```
class MyRunnable implements Runnable {
   String message;
   int delay;

   MyRunnable(String msg, int d) {
      message = msg;
      delay = d;
   }
   public void run() {
      try {
         while (true) {
            System.out.println(message);
            Thread.sleep(delay);
         }
      } catch (InterruptedException e) {
         System.out.println(e);
      }
   }
}
public class MultiThreadDemo {
   public static void main(String[] args) {
      new Thread(new MyRunnable("Good Morning", 1000)).start();
      new Thread(new MyRunnable("Hello", 2000)).start();
      new Thread(new MyRunnable("Welcome", 3000)).start();
   }
}
```

OUTPUT

```
Good Morning
Hello
Welcome
Good Morning
Good Morning
Hello
Good Morning
Welcome
...
```

7b.write a java program that correctly implements producer consumer problem using the concept of interthread communication

```
class SharedBuffer {
    private int data;
    private boolean hasData = false;

    // Produce method
    synchronized void produce(int value) {
        while (hasData) {
            try {
                wait(); // wait if data is not yet consumed
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
        data = value;
        System.out.println("Produced: " + data);
        hasData = true;
        notify(); // notify consumer
    }

    // Consume method
    synchronized int consume() {
        while (!hasData) {
            try {
                wait(); // wait if no data available
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
        System.out.println("Consumed: " + data);
        hasData = false;
        notify(); // notify producer
        return data;
    }
}
```

```
// Producer Thread
class Producer extends Thread {
    SharedBuffer buffer;

    Producer(SharedBuffer buffer) {
        this.buffer = buffer;
    }

    public void run() {
        for (int i = 1; i <= 5; i++) {
            buffer.produce(i);
        }
    }
}

// Consumer Thread
class Consumer extends Thread {
    SharedBuffer buffer;

    Consumer(SharedBuffer buffer) {
        this.buffer = buffer;
    }

    public void run() {
        for (int i = 1; i <= 5; i++) {
            buffer.consume();
        }
    }
}

// Main class
public class ProducerConsumerDemo {
    public static void main(String[] args) {
        SharedBuffer buffer = new SharedBuffer();

        Producer p = new Producer(buffer);
        Consumer c = new Consumer(buffer);
```

```
        p.start();
        c.start();
    }
}
```

OUTPUT:

Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5

Experiment-8

Write a Java program that works as asimple calculator. Use a grid layout to arrange button for the digits and for the +, -,*, % operations. Add a text field to display the result

```
package calc;
import javax.swing.*;
import
javax.swing.event.*;
import java.awt.*;
import java.awt.event.*;
class A extends JFrame implements ActionListener {
 public JButton b1, b2, b3, b4, b5, b6, b7, b8, b9, b10, b11, b12, b13, b14, b15, b16;
 public JTextField tf1;
 public JPanel p;
 public String v = "";
 public String v1 =
 "0"; public String op
 = ""; public A() {
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setSize(400, 400);
 p = new JPanel(new
FlowLayout()); tf1 = new
JTextField(10);  p.add(tf1);
 add(p);
setLayout(new GridLayout(0, 3));
 b1 = new JButton("1");
 b1.addActionListener(this);
 add(b1);
 b2 = new JButton("2");
 b2.addActionListener(this
 ); add(b2);
 b3 = new JButton("3");
 b3.addActionListener(this
 ); add(b3);
 b4 = new JButton("4");
 b4.addActionListener(this
 ); add(b4);
 b5 = new JButton("5");
 b5.addActionListener(this
 ); add(b5);
 b6 = new JButton("6");
 b6.addActionListener(this
 ); add(b6);
```

```
 b7 = new JButton("7");
 b7.addActionListener(this);
add(b7);
 b8 = new JButton("8");
 b8.addActionListener(this
 ); add(b8);
 b9 = new JButton("9");
 b9.addActionListener(this
 ); add(b9);
 b10 = new JButton("0");
 b10.addActionListener(this);
 add(b10);
 b11 = new JButton("+");
 b11.addActionListener(this);
 add(b11);
 b12 = new JButton("-");
 b12.addActionListener(this);
 add(b12);
 b13 = new JButton("*");
 b13.addActionListener(this);
 add(b13);
 b14 = new JButton("/");
 b14.addActionListener(this);
 add(b14);
 b16 = new JButton("%");
 b16.addActionListener(this);
 add(b16);
 b15 = new JButton("=");
 b15.addActionListener(this);
 add(b15);
setVisible(true);
}
public void actionPerformed(ActionEvent ae) {
String b = ae.getActionCommand();
switch (b) {
case "1": {
v = v + "1";
tf1.setText(v);
}
break;
case "2":
{
v = v + "2";
tf1.setText(v);
}
break;
case "3":
{
```

```
 v = v + "3";
tf1.setText(v)
}
 break;
 case "4": {
 v = v + "4";
 tf1.setText(v);
 }
 break;
 case "5": {
 v = v + "5";
 tf1.setText(v);
 }
 break;
 case "6": {
 v = v + "6";
 tf1.setText(v);
 }
 break;
 case "7": {
 v = v + "7";
 tf1.setText(v);
 }
 break;
 case "8": {
 v = v + "8";
 tf1.setText(v);
 }
 break;
 case "9": {
 v = v + "9";
 tf1.setText(v);
 }
 break;
 case "0": {
 v = v + "0";
 tf1.setText(v);
 }
 break;
 case "+": {
 op = "+";
 v1 = tf1.getText();
 v = "";
 }
 break;
 case "-": {
 op = "-";
```

```
v1 = tf1.getText();
v = "";
 }
 break;
 case "*": {
 op = "*";
 v1 = tf1.getText();
 v = "";
 }
 break;
 case "/": {
 op = "/";
 v1 = tf1.getText();
 v = "";
 }
 break;
 case "%": {
 op = "%";
 v1 = tf1.getText();
 v = "";
 }
 break;
 case "=": {
 switch (op) {
 case "+": {
 v = tf1.getText();
 if (v.equals("")) {
 v = "0";
 }
 long i = Long.parseLong(v1) + Long.parseLong(v);
 tf1.setText(String.valueOf(i));
 v="";
 }
 break;
 case "-": {
 v = tf1.getText();
 if (v.equals("")) {
 v = "0";
 }
 long i = Long.parseLong(v1) - Long.parseLong(v);
 tf1.setText(String.valueOf(i));
 v="";
 }
 break;
```

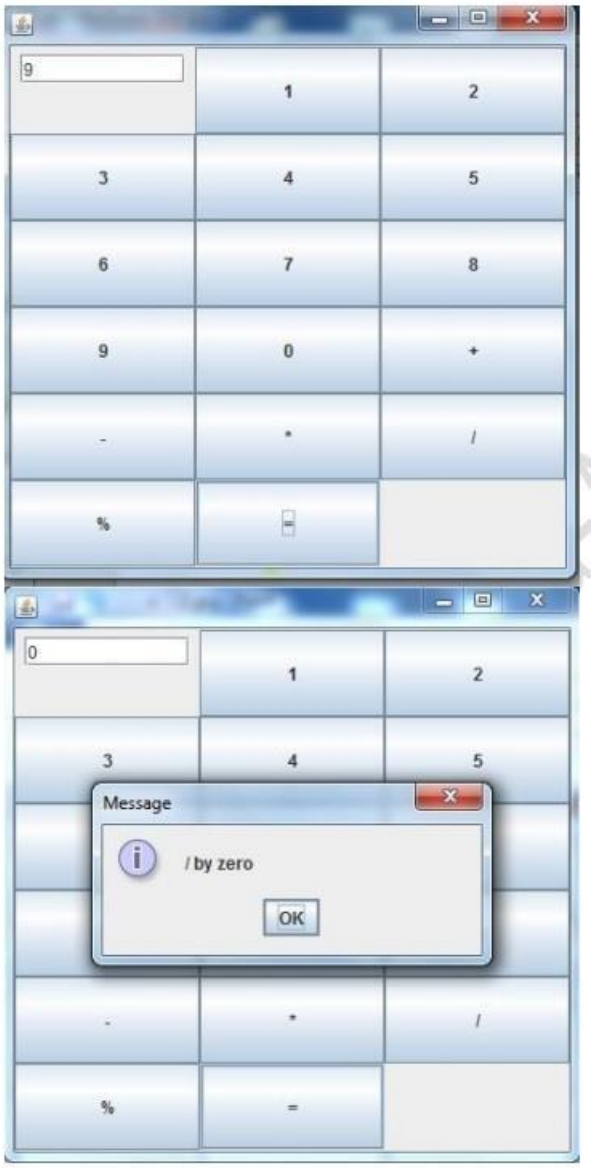
```
 case "*":

{

 v = tf1.getText();
if (v.equals("")) {
v = "0";
 }
 long i = Long.parseLong(v1) * Long.parseLong(v);
 tf1.setText(String.valueOf(i));
 v="";
 }
 break;
 case "/": {
 try {
 v = tf1.getText();
 if (v.equals("")) {
 v = "0";
 }
 long i = Long.parseLong(v1) / Long.parseLong(v);
 tf1.setText(String.valueOf(i));
 v="";
 } catch (Exception ex) {
JOptionPane.showMessageDialog(this, ex.getMessage());
 }
 }
 break;
 case "%": {
 try {
 v = tf1.getText();
 if (v.equals("")) {
 v = "0";
 }
 long i = Long.parseLong(v1) % Long.parseLong(v);
 tf1.setText(String.valueOf(i));
 v="";
 } catch (Exception ex) {
JOptionPane.showMessageDialog(this, ex.getMessage());
 }
 }
 break;
 }
 }
 break;
 }
 }
}
public class Calc {
 public static void main(String[] args) {
```

A a = new A();} }

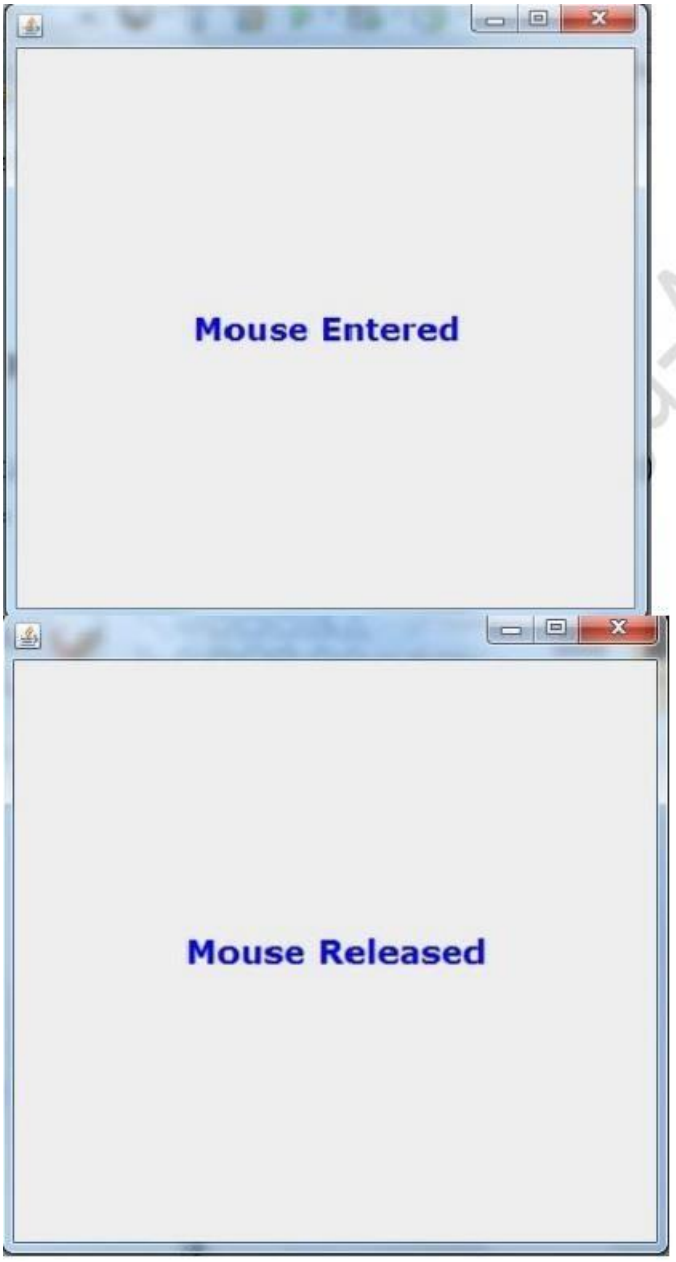
output



EXPERIMENT-9

a) **Write a Java program that handles all mouse events and shows the event name at the center of the window when a mouse event is fixed (Use Adapter Classes).**

```
import javax.swing.*;
import java.awt.*;
import javax.swing.event.*;
import java.awt.event.*;
class A extends JFrame implements MouseListener {
JLabel l1;
 public A() {
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setSize(400, 400);
setLayout(new GridBagLayout());
 l1 = new JLabel();
 Font f = new Font("Verdana", Font.BOLD, 20);
 l1.setFont(f);
 l1.setForeground(Color.BLUE);
 l1.setAlignmentX(Component.CENTER_ALIGNMENT);
 l1.setAlignmentY(Component.CENTER_ALIGNMENT);
 add(l1);
addMouseListener(this);
setVisible(true);
 }
 public void mouseExited(MouseEvent m) {
 l1.setText("Mouse Exited");
 }
 public void mouseEntered(MouseEvent m) {
 l1.setText("Mouse Entered");
 }
 public void mouseReleased(MouseEvent m) {
 l1.setText("Mouse Released");
 }
 public void mousePressed(MouseEvent m) {
 l1.setText("Mouse Pressed");
 }
 public void mouseClicked(MouseEvent m) {
 l1.setText("Mouse Clicked");
 }
}
public class Mevents {
 public static void main(String[] args) {
 A a = new A();
 }
```


Mouse Entered
Mouse Released

b) write a java program for handling key events.

```
import java.awt.*;
import java.awt.event.*;
class KeyEventDemo extends Frame implements KeyListener
{
   Label l;
   KeyEventDemo() {
      setTitle("Key Event Demo");
      setSize(400, 200);
      setLayout(new FlowLayout());
      l = new Label("Press any key...");
      add(l);
      addKeyListener(this);
      setVisible(true);
      // Close window
      addWindowListener(new WindowAdapter() {
         public void windowClosing(WindowEvent we) {
            dispose();
         }
      });
   }
   // When key is pressed
   public void keyPressed(KeyEvent e) {
      l.setText("Key Pressed: " + e.getKeyChar());
   }
   // When key is released
   public void keyReleased(KeyEvent e) {
      l.setText("Key Released: " + e.getKeyChar());
   }

   // When key is typed
   public void keyTyped(KeyEvent e) {
      l.setText("Key Typed: " + e.getKeyChar());
   }

   public static void main(String[] args) {
      new KeyEventDemo();
   }
}
```

Output:

A window opens with message: **"Press any key..."**
When you press a key:
Displays **Key Pressed: A**

When you release:
Displays **Key Released: A**

When typed:
Displays **Key Typed: A**

EXPERIMENT-10

Write a program that creates a user interface to perform integer divisions. The user enters two numbers in the text fields Num1 and Num 2.

The division of Num1 and Num2 is displayed in the Result field when the Divide button is clicked. If Num1or Num2 were not an integer, the program would throw Number Format Exception. If Num2 were Zero, the program would throw an Arithmetic Exception and display the exception in a message dialog box.

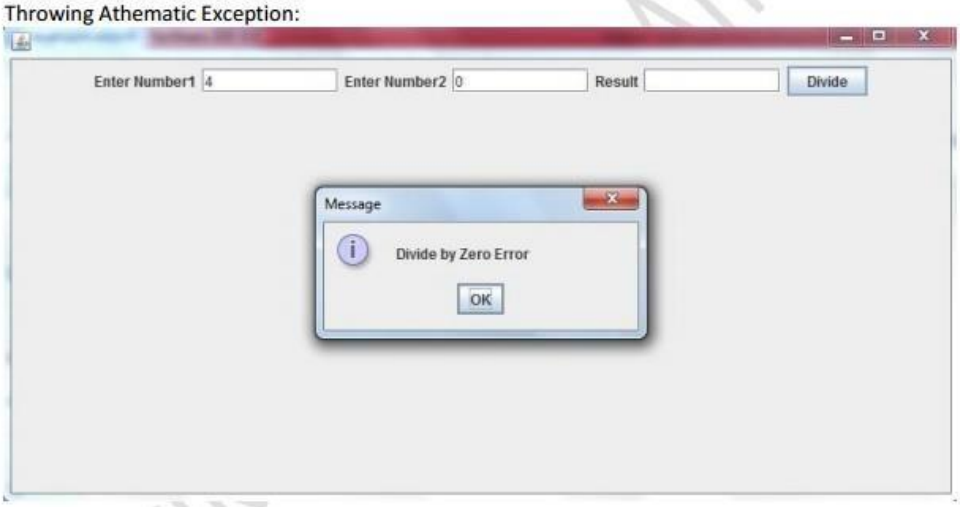
```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import javax.swing.event.*;
class A extends JFrame implements ActionListener {
JLabel l1, l2, l3;
JTextField tf1, tf2, tf3;
JButton b1;
A() {
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new FlowLayout());
 l1 = new JLabel("Welcome");
setSize(800, 400);
 l1 = new JLabel("Enter Number1");
 add(l1);
 tf1 = new JTextField(10);
 add(tf1);
 l2 = new JLabel("Enter Number2");
 add(l2);
 tf2 = new JTextField(10);
 add(tf2);
 l3 = new JLabel("Result");
 add(l3);
 tf3 = new JTextField(10);
 add(tf3);
 b1 = new JButton("Divide");
add(b1);
b1.addActionListener(this);
setVisible(true);
 }
 public void actionPerformed(ActionEvent ae) {
 try {
 int a = Integer.parseInt(tf1.getText());
int b = Integer.parseInt(tf2.getText());
if(b==0)
 throw new ArithmeticException(" Divide by Zero Error");
```

```
float c = (float) a / b;
 tf3.setText(String.valueOf(c));
 } catch (NumberFormatException ex) {
JOptionPane.showMessageDialog(this, ex.getMessage());
 } catch (ArithmeticException ex) {
JOptionPane.showMessageDialog(this, ex.getMessage());
 }
 }
}
public class JavaApplication9 {
 public static void main(String[] args) {
 A a = new A();
 }
}
```

Correct Output:

Throwing Number Format Exception:

Throwing Athematic Exception:



EXPERIMENT-11

a) Write a Java program that simulates a traffic light. The program lets the user select one of three lights: red, yellow or green with radio buttons. On selecting a button, an appropriate message with “Stop” or “Ready” or “Go” should appear above the buttons in selected color. Initially, there is no message shown.

```
import javax.swing.*;
import javax.swing.event.*;
import java.awt.*;
import java.awt.event.*;
class A extends JFrame implements ItemListener {
 public JLabel l1, l2;
 public JRadioButton r1, r2, r3;
 public ButtonGroupbg;
 public JPanel p, p1;
 public A() {
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new GridLayout(2, 1));
setSize(800, 400);
 p = new JPanel(new FlowLayout());
 p1 = new JPanel(new FlowLayout());
 l1 = new JLabel();
 Font f = new Font("Verdana", Font.BOLD, 60);
 l1.setFont(f);
 add(l1);
p.add(l1);
 add(p);
 l2 = new JLabel("Select Lights");
 p1.add(l2);
JRadioButton r1 = new JRadioButton("Red Light");
 r1.setBackground(Color.red);
 p1.add(r1);
 r1.addItemListener(this);
JRadioButton r2 = new JRadioButton("Yellow Light");
 r2.setBackground(Color.YELLOW);
 p1.add(r2);
 r2.addItemListener(this);
JRadioButton r3 = new JRadioButton("Green Light");
 r3.setBackground(Color.GREEN);
 p1.add(r3);
 r3.addItemListener(this);
 add(p1);
bg = new ButtonGroup();
bg.add(r1);
bg.add(r2);
bg.add(r3);
```

```
setVisible(true);
 }
 public void itemStateChanged(ItemEventi) {
JRadioButtonjb = (JRadioButton) i.getSource();
switch (jb.getText()) {
 case "Red Light": {
 l1.setText("STOP");
 l1.setForeground(Color.red);
 }
 break;
 case "Yellow Light": {
 l1.setText("Ready");
 l1.setForeground(Color.YELLOW);
 }
 break;
 case "Green Light": {
 l1.setText("GO");
 l1.setForeground(Color.GREEN);
 }
 break;
 }
 }
}
public class TLights {
 public static void main(String[] args) {
 A a = new A();
 }
}
```

Output:

STOP
Select Lights
Yellow Light
Green Light

Ready
Select Lights
Yellow Light
Green Light

GO
Select Lights
Yellow Light
Green Light

b) Write a Java program that allows the user to draw lines, rectangles and ovals.

```
import java.awt.*;
import java.awt.event.*;

class DrawShapes extends Frame implements MouseListener {
    int x1, y1, x2, y2;
    String shape = "Line";

    DrawShapes() {
        setTitle("Draw Shapes");
        setSize(500, 500);
        setLayout(new FlowLayout());

        // Buttons
        Button lineBtn = new Button("Line");
        Button rectBtn = new Button("Rectangle");
        Button ovalBtn = new Button("Oval");

        add(lineBtn);
        add(rectBtn);
        add(ovalBtn);

        // Button actions
        lineBtn.addActionListener(e -> shape = "Line");
        rectBtn.addActionListener(e -> shape = "Rectangle");
        ovalBtn.addActionListener(e -> shape = "Oval");

        addMouseListener(this);

        setVisible(true);

        // Close window
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                dispose();
            }
        });
```

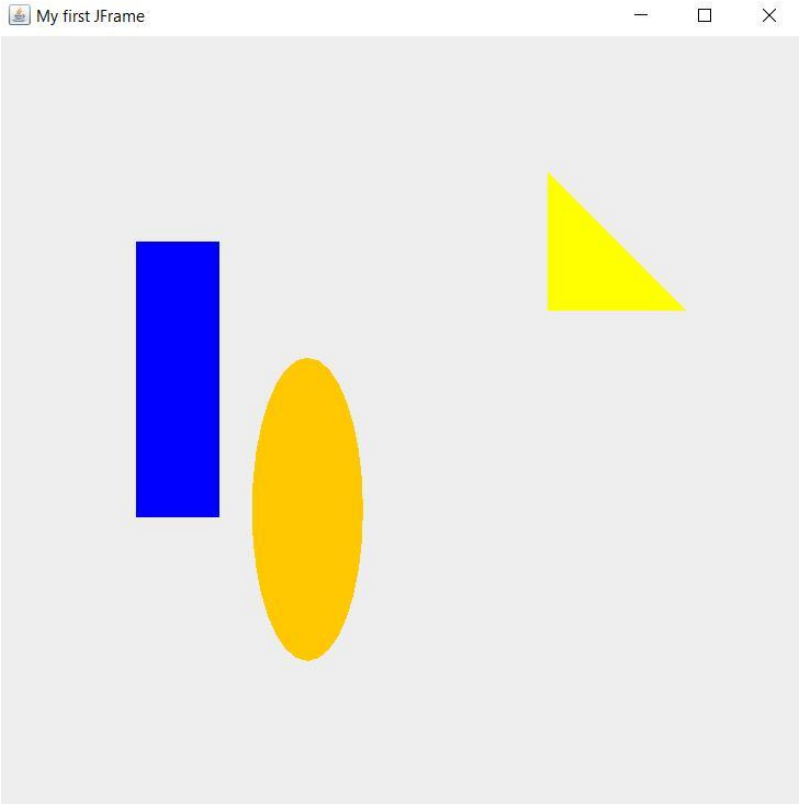
```
    }
    public void mousePressed(MouseEvent me) {
        x1 = me.getX();
        y1 = me.getY();
    }
    public void mouseReleased(MouseEvent me) {
        x2 = me.getX();
        y2 = me.getY();
        repaint();
    }
    public void paint(Graphics g) {
        g.setColor(Color.BLUE);

        int width = Math.abs(x2 - x1);
        int height = Math.abs(y2 - y1);
        int x = Math.min(x1, x2);
        int y = Math.min(y1, y2);

        if (shape.equals("Line")) {
            g.drawLine(x1, y1, x2, y2);
        } else if (shape.equals("Rectangle")) {
            g.drawRect(x, y, width, height);
        } else if (shape.equals("Oval")) {
            g.drawOval(x, y, width, height);
        }
    }

    // Unused methods of MouseListener
    public void mouseClicked(MouseEvent me) {}
    public void mouseEntered(MouseEvent me) {}
    public void mouseExited(MouseEvent me) {}
}
public class ShapeDrawingApp {
    public static void main(String[] args) {
        new DrawShapes();
    }
}
```

OUTPUT:



EXPERIMENT-12

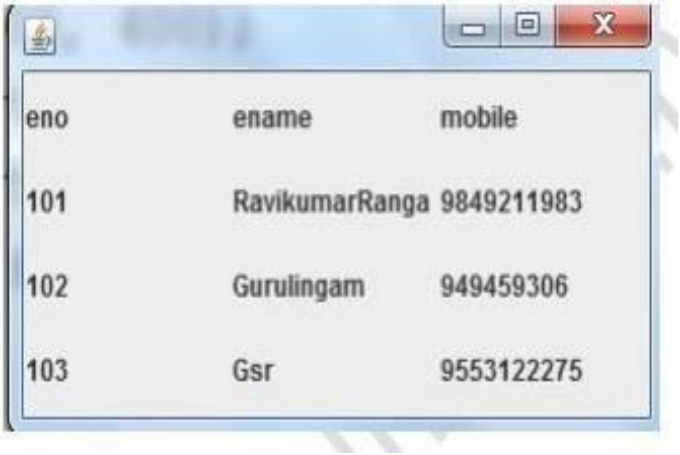
Create a table in Table.txt file such that the first line in the file is the header, and the remaining lines correspond to rows in the table. The elements are separated by commas. Write a java program to display the table using JTable component

```
import java.io.*;
import java.util.*;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import javax.swing.event.*;
class Text_To_Table extends JFrame
{
   public void convertTexttotable()
   {
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setSize(400,300);
GridLayout g = new GridLayout(0, 4);
setLayout(g);
       try
       {
FileInputStreamfis = new FileInputStream("./Table.txt");
           Scanner sc = new Scanner(fis);
String[] arrayList;
           String str;
           while (sc.hasNextLine())
           {
               str = sc.nextLine();
arrayList = str.split(",");
               for (String i : arrayList)
               {
add(new Label(i));
               }
           }
       }
       catch (Exception ex) {
ex.printStackTrace();
       }
setVisible(true);
setTitle("Display Data in Table");
   }
}
public class TableText
{
   public static void main(String[] args)
```

```
    {
Text_To_Tablett = new Text_To_Table();
tt.convertTexttotable();
    }
}
```

Output:

eno,ename,mobile



101,RavikumarRanga,9849211983
102,Gurulingam,949459306
103,Gsr,9553122275